

Information-Set Conditional Integrity Auditing for Hybrid Quantum-Classical Kernel Workflows

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Abstract

Integrity auditing asks whether evidence exposed at a workflow boundary can identify a consequential intervention. We formalize **information-set conditional auditability** and link interventions, evidence regimes, sensor families, blind regions and responses. For a fixed deterministic predictor, evaluation-label changes leave feature and prediction statistics invariant; if class counts are preserved, label-marginal statistics are invariant too. These elementary results support an integrated coverage methodology rather than a standalone theorem claim.

We evaluate an RBF support-vector classifier and three empirically distinct fidelity-kernel profiles under 18 controlled interventions. A frozen CICIDS2017 gate and 360 exact-statevector expansion jobs cover CICIDS2017, UNSW-NB15 and ToN-IoT, including scale and five fixed out-of-distribution settings. After verified deduplication, the expansion contains 11,400 model-intervention observations. All 3,600 evaluation-label observations leave features, scores and predictions unchanged; all 1,800 prior-preserving cases also leave label marginals unchanged. Every one of the 2,184 positive conclusion impacts changes item-aligned joint-outcome evidence.

A separate 165-cell simulator gate shows complementary quantum-workflow coverage: semantic controls prevent hash-only false alarms, parameterized circuit changes are exposed, and algebraic checks miss positive-semidefinite kernel substitution. Controlled finite-shot emulators exercise repeated estimation. A four-contract demonstrator converts the coverage map into hash-chained, fail-closed **allow/hold/block** decisions across six scenarios. The contribution is the combined formalization, coverage validation, bounded quantum layer and executable response. Heterogeneous ZZ-versus-SVC effects are secondary. No hardware, scheduling, provider, multi-tenancy or operational Fleet Management claim is made.

Keywords: hybrid quantum-classical systems; quantum machine learning; quantum kernels; integrity auditing; observability; sensor coverage; cybersecurity

1. Introduction

Hybrid quantum-classical software distributes one reported result across several computational and evidential boundaries. Classical records are acquired, cleaned, projected and encoded; a circuit and execution stack produce quantum evidence; a classical learner converts a kernel or measurement into a decision; and an evaluation layer aligns that decision with reference outcomes and reports a metric. A plausible final number may therefore coexist with a failure in the data, circuit, kernel, prediction, label, or reporting path. Security analyses of hybrid systems accordingly call for lifecycle-wide controls rather than trust in an isolated algorithm [Volya2023, Ghosh2023].

Robustness benchmarking normally collapses this chain into a performance change. That endpoint is necessary, but it does not identify what changed or whether the available evidence could reveal the change. A large performance loss can be easy to diagnose because several independent sensors react. A smaller loss may be a more serious integrity failure when it changes the reported conclusion while every sensor available to the evaluator remains invariant. Conversely, a drift statistic can react strongly even when the fixed decision is unchanged. Impact, observability and response are different axes.

The distinction matters especially at the evaluation boundary. Let a deployed predictor receive exactly the same feature batch and emit exactly the same predictions, but let the reference labels used to score that batch be corrupted. The reported balanced accuracy can change although the model has not degraded. A feature monitor cannot observe the event because its input is unchanged. A prediction monitor adds no information because the predictions are unchanged. A label-count monitor can expose an unbalanced corruption, but a balanced exchange of class identities leaves that marginal invariant. Detection then requires an item-aligned comparison between outcome and prediction, independently authenticated labels, or provenance that binds labels to items. Calling the event simply “stealthy” suppresses the crucial qualifier: stealthy to whom, with which evidence?

This paper makes that qualifier explicit. **Information-set conditional auditability** evaluates an intervention relative to the evidence exposed at a workflow boundary and to a declared sensor family. It separates changes in model output from changes in the evidence used to evaluate that output, then records the remaining blind region. Quantum kernels are a useful bounded demonstrator: their workflow crosses classical preprocessing, parameterized circuit construction, fidelity evaluation, a kernel matrix, a classical support-vector decision and an outcome-based evaluation. The setting is quantum-specific without requiring claims about a physical processor.

Three research questions organize the study:

- **RQ1 — conditional identifiability:** Which controlled interventions are observable from feature, prediction, label-marginal, item-aligned outcome and quantum-workflow evidence, and where are the exact blind regions?
- **RQ2 — transport of the coverage boundary:** Do the information-set blind regions recur across datasets, fixed ID/OOD environments, projected dimensions and classical/quantum kernel profiles?
- **RQ3 — quantum-boundary composition:** Do provenance, semantic, algebraic, repeated-estimation and output sensors provide complementary coverage, and can their evidence drive an executable fail-closed decision?

The contribution is one integrated package, not a claim that any constituent is individually new:

1. a workflow formalization that separates model-output, evaluation-conclusion and execution-integrity effects and conditions every auditability claim on an information set;
2. exact blind-region statements plus an attack-to-sensor coverage method that distinguishes structural invariance from low empirical detector power;
3. multi-dataset and fixed-OOD validation with verified computational-unit deduplication and uncertainty clustered at the data-split level;
4. a bounded quantum-specific layer combining circuit/kernel provenance, semantic comparison, algebraic validation, repeated estimation and output monitoring; and
5. an executable four-contract demonstrator whose **allow/hold/block** policy makes the declared coverage operational at a local research-prototype level.

The paper deliberately does **not** claim that Propositions 1–2 are deep new impossibility theorems, that contracts or hash chains originate here, or that a particular quantum feature map is universally less robust than an RBF kernel. The scope covers data-to-evaluation integrity and selected simulated quantum-kernel boundaries. Quantum hardware, calibrated backend noise, scheduling, provider-side security, multi-tenant interference and the project’s Fleet Management case study are outside the scientific claim.

2. Related work and positioning

2.1 Hybrid quantum-classical security and integrity

Volya et al. decompose classical-quantum systems into classical control, compilation, quantum execution and measurement surfaces, showing why security cannot be inferred from the circuit alone [Volya2023]. Surveys broaden this view to malicious compilation, hardware faults, cloud exposure, side channels and classical dependencies [Ghosh2023, Saki2022]. Quantum Vulnerability Factor quantifies changes in circuit output distributions under injected faults [Oliveira2024], and Quantum Leak demonstrates timing leakage in cloud quantum services [Lu2025]. These studies establish the threat surface and impact metrics; they do not make

auditability conditional on the evidence exposed to a sensor. Our empirical scope is narrower than their lifecycle, and our claim is about the coverage argument within that scope.

The closest direct overlap is QCIVET, a concurrent 2026 preprint that models a hybrid pipeline with explicit contracts, hash-chained audit traces and semantic observable checks grounded in behavioural subtyping [YeniarasKarimov2026]. It also includes calibrated-noise and real-QPU evidence that this paper does not possess. Consequently, stage contracts, hash chains and fail-closed verification are not claimed here as standalone novelty. The distinction is the unit of analysis: we ask which *information set* makes a changed asset identifiable, treat evaluation labels and conclusions as first-class boundaries, validate an attack-to-sensor coverage map across multiple data environments, and use the contract only to execute that map.

A second concurrent line evaluates circuit integrity with complementary structural, interaction and behavioural scores [Ahmed2026]. It reinforces the case against a one-metric integrity claim. Our quantum gate reaches the same general lesson in a kernel workflow but embeds it in a wider evidence lattice that includes classical features, predictions, item-aligned labels and the reported benchmark conclusion. We therefore do not claim that complementarity among circuit metrics is itself new.

2.2 Quantum software testing, verification and contracts

Quantum software testing addresses the oracle problem through cross-stack or semantics-preserving variants. QDiff uses differential testing of quantum software stacks [Wang2021]; MorphQ generates quantum-specific metamorphic relations for Qiskit and revealed platform bugs [Paltenghi2023]; and QEMI extends equivalence-modulo-inputs testing to Qiskit, Q# and Cirq [Luo2026]. Formal and runtime approaches include CertiQ compiler verification [Shi2019], quantum program logics [Ying2011, Zhou2019], projection-based runtime assertions [Li2020], and design-by-contract proposals for quantum software [Yamaguchi2023].

These methods verify programs, compilers or semantic relations. The benign transpilation and common-unitary conditions in our simulator gate are therefore negative controls, not a claim to have invented metamorphic testing. Our target is different: to expose what a provenance hash, semantic probe, algebraic check, repeated estimate or final prediction can and cannot establish about one hybrid kernel workflow. The result is a sensor-coverage contract rather than a general quantum compiler or platform tester.

2.3 Monitoring under incomplete or delayed labels

Classical monitoring work already makes clear that labels alter what can be inferred. Black-box shift estimation detects and corrects label shift under assumptions on the relationship between source and target distributions [Lipton2018]. MLDemon combines unlabeled observations with selectively acquired labels to monitor deployed performance [Ginart2022], and recent uncertainty-aware monitoring likewise uses active labeling intervention [Koebler2025]. A concurrent evidence-sufficiency preprint studies delayed ground truth, proxy coverage and the inability of feature proxies to detect pure concept drift when the feature distribution is unchanged [Solozobov2026].

Our intervention should not be confused with statistical label shift. It changes the *evaluation labels attached to fixed items* while holding the input and predictor fixed; it is a controlled integrity failure in the scoring path. Moreover, the label source may itself be compromised, so label availability is not enough: the audit must distinguish a trusted item-aligned outcome from an unauthenticated marginal. The contribution is thus not the general observation that monitoring without labels is limited, but its integration into an explicit hybrid-workflow coverage and response contract.

2.4 Benchmark validity, drift sensors and quantum kernels

MMD supplies a kernel two-sample statistic [Gretton2012], while KS summaries and Jensen-Shannon divergence provide familiar univariate and histogram views. A non-zero raw statistic is not automatically a calibrated alarm: thresholds, false-alarm control and power belong to a deployment-specific test. This paper therefore uses exact equality only for constructed invariances and describes non-invariant sensor magnitudes without treating them as operating points.

Security-oriented ML benchmarks are vulnerable to sampling, labeling, baseline, metric and threat-model errors [Arp2022]. Quantum-kernel benchmarks add strong dependence on feature map, data encoding, pre-processing and classical comparator [Schnabel2025]. Work on adversarial quantum kernels and QML poisoning asks whether an attack changes performance and whether a defense recovers it [Montalbano2025, Kundu2024]. Our secondary ZZ-versus-SVC profile belongs in this benchmarking context. It is retained to show that impact profiles vary by model, not to infer a universal quantum/classical ordering.

2.5 Resulting gap

Table 1 summarizes the nearest-work comparison. In the search reported in the supplementary novelty audit, we found no prior study combining the same five elements: information-set conditioning; labels and the evaluation conclusion as explicit integrity boundaries; exact and empirical sensor blind regions; multi-dataset/fixed-OOD validation in quantum-kernel workflows; and an executable fail-closed instantiation. The novelty is this combination. Removing the conditioning would reduce the paper to another detector benchmark; removing the quantum layer would lose the hybrid boundary; and presenting the contract or the two propositions alone would overlap established work.

Line of work	Evidence conditioning	Evaluation-label boundary	Quantum evidence	Validation and response
Hybrid threat taxonomies [Volya2023, Ghosh2023]	none	no	broad lifecycle	taxonomy; no response
QCIVET [YeniarasKari-mov2026]	stage observables/contracts	no	calibrated noise and QPU	three applications; verification engine
MorphQ/QDiff/QEM [Wang2021, Paltenghi2023, Luo2026]	semantic relations	no	software stack	cross-platform test verdicts
Evidence sufficiency [Solozobov2026]	proxy coverage	delayed outcomes	no	one dataset; readiness gate
Circuit-integrity metrics [Ahmed2026]	fixed metric layers	no	circuit metrics	circuit corpus; no response
This work	explicit evidence partial order	yes	bounded simulator	3 datasets; 8 environments; local fail-closed response

Table 1. Positioning against the closest lines of work. “Partial” does not denote inferiority; it denotes a different formal object. QCIVET includes QPU evidence beyond the scope of this paper.

3. Formal framework

3.1 Workflow and impact loci

Let a hybrid evaluation workflow be represented by

$$\mathcal{W} = (X, P, C, K, E, f, y, R),$$

where $\mathbf{X}$ is the acquired feature batch, $\mathbf{P}$ is classical preprocessing, $\mathbf{C}$ is the parameterized circuit or feature map, $\mathbf{K}$ is the estimated kernel, $\mathbf{E}$ is execution metadata, $\mathbf{f}$ is the resulting predictor, $\mathbf{y}$ is reference outcome information, and $\mathbf{R}$ is the reported evaluation result.

A perturbation $\mathbf{a}$ may act on one or more elements of $\mathbf{W}$. We distinguish:

- **model-output impact**, measured by a change in predictions or scores on aligned items;
- **benchmark-conclusion impact**, measured by a change in a reported metric or comparative conclusion;
- **execution-integrity impact**, measured by a change in circuit, kernel, or execution evidence even when the final prediction is unchanged.

The current primary performance endpoint is the balanced-accuracy conclusion impact

$$I_R(a) = BA_{clean} - BA_a.$$

For aligned deterministic predictions, output impact includes

$$I_f(a) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{f_a(x_i) \neq f_0(x_i)\}.$$

When $I_R(a) > 0$ but $I_f(a) = 0$, the measured change lies in the evaluation or label path, not in the predictor output.

3.2 Information regimes

We use a partially ordered family of evidence sets rather than a falsely linear hierarchy:

- I_X : features or preprocessed features only;
- I_{XF} : features plus model scores or predictions, with $I_X \subset I_{XF}$;
- I_{Ym} : the evaluation-label marginal or class-count vector;
- I_{XFY} : item-aligned features, predictions and reference outcomes, which contains both I_{XF} and the outcome marginal;
- I_Q : circuit, transpilation, kernel-estimation and execution evidence.

I_{Ym} is not an intermediate step between I_{XF} and I_{XFY} , and I_Q is a separate workflow branch that can be joined with classical evidence. A sensor $\mathbf{S}$ is admissible in regime $\mathbf{I}$ if it is a measurable function only of evidence in $\mathbf{I}$. Auditability is therefore reported as $\mathbf{A}(\mathbf{a} \mid \mathbf{I}, \mathbf{S})$, never as an unqualified property of $\mathbf{a}$.

3.3 Exact observability boundaries

Proposition 1 (feature and prediction blindness to label-only perturbations). Let $\mathbf{f}$ be fixed and deterministic. Let T_y transform only the evaluation labels, so that $\mathbf{X}' = \mathbf{X}$ and $\mathbf{f}'(\mathbf{X}') = \mathbf{f}(\mathbf{X})$. Then every audit statistic measurable with respect to I_X or I_{XF} is invariant under T_y .

Proof. The complete arguments supplied to such a statistic are identical before and after the intervention. A deterministic function evaluated on identical arguments returns the same value. Therefore no test based only on those arguments can distinguish the two states. $\square$

Proposition 2 (label-marginal blindness under prior preservation). If T_y additionally preserves the empirical class histogram, every audit statistic measurable only with respect to I_{Ym} is invariant.

Proof. The sufficient input to the statistic, the empirical label-count vector, is unchanged. $\square$

Corollary 1. Detecting a prior-preserving label-only perturbation requires at least one source outside I_X , I_{XF} , and the label histogram: joint label-prediction evidence, item-level provenance, an independently authenticated label source, or an assumption restricting admissible label transformations.

These propositions are finite-batch statements and require no distributional asymptotics. They describe sensor capability, not attack prevalence.

3.4 Conditional auditability gap

For an information regime $\mathbf{I}$, sensor family $\mathbf{S}$, and materiality threshold τ , an intervention belongs to the auditability-gap set if

$$\mathcal{G}_{I,S}(\tau) = \{a : |I_R(a)| \geq \tau \wedge S(a) = S(0)\}.$$

The absolute endpoint covers both degradation and apparent improvement caused by an integrity failure; the present positive-impact counts use the signed drop for interpretability. Exact invariance is used only where the intervention supplies it. For non-invariant signals, practical detection requires a separately calibrated alarm rule and false-alarm budget. A min-max-normalized stealth score may summarize a fixed experiment, but it is neither the definition of an auditability gap nor comparable outside its frozen normalization reference.

4. Methodology

4.1 Claim hierarchy and threat model

The primary estimand is sensor coverage conditional on an evidence set. Exact blindness under constructed label-path interventions is Tier A evidence. The multi-dataset and fixed-OOD replication of that coverage boundary is Tier B. Model-profile differences, including ZZ versus SVC, are Tier C descriptive evidence and cannot override the primary result.

The study is an integrity stress test. Its manipulations are controlled interventions, not claims about attack prevalence or operational adversary frequency. The data-to-evaluation gates trust the training routine and fitted model artifact while changing evaluation features, post-projection values, or labels. The quantum-specific simulator gate additionally permits circuit/parameter mutation and post-estimation kernel corruption. Malicious scheduling, calibrated backend tampering, multi-tenant interference, provider-side manipulation, side channels and physical-device execution are outside the empirical scope.

4.2 Data and fixed environments

Gate 1 uses a balanced binary subset derived from CICIDS2017 [Sharafaldin2018] with 3,000 rows and 77 numeric feature columns before projection and caps training/evaluation at 128/128. The expansion adds a prespecified 256/256 CICIDS scale gate; balanced ID subsets from UNSW-NB15 [Moustafa2015] and ToN-IoT [Alsaedi2020]; four temporal CICIDS train-test pairs; and one UNSW temporal pair. Expansion jobs use 128/128 except for the scale gate. All data-to-evaluation gates report projected dimensions 8, 10 and 12.

The eight expansion environments are CICIDS2017 ID at 256/256; UNSW-NB15 and ToN-IoT ID; CICIDS Tuesday-to-Wednesday, Tuesday-to-Friday-PortScan, Wednesday-to-Thursday-Web and Wednesday-to-Friday-AM temporal pairs; and one UNSW-NB15 temporal pair. The two additional ID datasets retain all quantum profiles; the scale and temporal gates compare SVC with ZZ. “OOD” identifies a fixed source-to-target construction, not a sampled deployment population.

These are fixed benchmark environments, not a probability sample of deployments. Each staged table is balanced at 3,000 rows; this is useful for a controlled audit but does not reproduce operational class prevalence. Categorical fields are excluded. Temporal OOD pairs are interpreted separately because several clean baselines approach chance performance, constraining the impact that subsequent interventions can express.

4.3 Preprocessing and models

Numeric non-finite values are replaced during staging, and train-defined preprocessing is applied without using evaluation labels. Each pipeline uses TruncatedSVD fitted on its training features. The RBF SVC

branch then applies a training-fitted standardization; the fidelity-kernel branch applies its frozen training-fitted angular scaling to $[0, 2\pi]$. Interventions are applied to the resulting evaluation representation. No hyperparameter search is performed. Accordingly, SVC/QSVC contrasts compare complete frozen pipelines and do not identify an isolated causal effect of kernel geometry.

The classical comparator is an RBF SVC. Quantum models use fidelity quantum kernels with `reps=1`. Four configured feature-map labels reduce to three unique empirical profiles because Z and PauliXZ generate identical kernel matrices under the frozen implementation: ZZ, PauliXYZ, and Z/PauliXZ-equivalent. The analysis therefore does not count Z and PauliXZ as independent behaviours.

Gate 1 used Qiskit Machine Learning’s reference ideal evaluator [JavadiAbhari2024]. Expansion gates use an explicitly tagged exact-statevector evaluator that prepares each unique feature-map state once and computes $K(x, y) = |\langle \phi(x) | \phi(y) \rangle|^2$ by matrix multiplication, with the same PSD repair for symmetric blocks. Unit matrices match the reference at $1\text{e-}10$; an end-to-end 128/128 replay preserves every discrete endpoint and integrity signal, with a maximum 0.0001222 difference confined to ROC-AUC quantities (one half-tie increment). This engine is an ideal-simulation optimization and supplies no shot-noise or QPU evidence.

4.4 Interventions

The 18 non-clean interventions comprise three frozen severities for each of six mechanisms:

- feature sign flip;
- feature-wise mean shift;
- scaling drift;
- feature dropout;
- random label flip;
- prior-preserving label flip.

The first four may change model input and output. The last two keep features, scores and predictions fixed and act only on the evaluation path. The code and archived tables retain the historical family label `target_shift`; in the paper these rows are called *evaluation-label interventions*, because they are not the distributional label-shift setting of [Lipton2018]. Random flips may alter the observed label marginal; prior-preserving exchanges do not. All six mechanisms receive equal weight in the suite average. This is a transparent engineering profile, not an estimate of real-world threat frequency.

4.5 Sensor families

Feature-only evidence uses feature JSD, MMD, and the KS rejection rate. Prediction-aware evidence adds score-distribution JSD. Label-marginal evidence uses prior shift and label JSD. Joint-outcome evidence includes prediction-rate, disagreement, prediction-distribution, and confusion-profile changes.

Sensor-family results are primary. Shared min-max normalization is retained only for within-design descriptive summaries. Raw invariance is used for structural-blindness claims.

4.6 Experimental units, endpoints and uncertainty

Gate 1 crosses five split seeds {42, 43, 44, 45, 46} with four nested model/perturbation seeds {42, 43, 44, 45}. The raw grid contains 7,980 rows. Because every quantum-map CSV repeats the same classical pipeline, 3,420 SVC rows are exact computational repetitions. Equality is verified across performance, impact, and signal columns before deduplication, leaving 4,560 unique model-attack observations.

The eight expansion environments cross the same five split seeds with two nested model seeds {42, 43}. All 360 expected CSV/JSON job pairs completed. Their 13,680 raw rows contain 2,280 verified repeated SVC rows from the multi-map ID gates, leaving 11,400 unique observations. No repeated key is collapsed unless performance, impact, and raw sensor evidence agree numerically.

The primary coverage endpoints are exact raw invariance/response counts within the prespecified design. The secondary model-profile endpoint is the within-split difference in suite-average balanced-accuracy con-

clusion impact. The split seed is its inferential unit: model-seed values are averaged within split, giving five clusters. Two-sided 95% Student-t intervals use four degrees of freedom. They describe resampling uncertainty inside each fixed environment; they are not population intervals and are not adjusted for simultaneous testing. Analyses over all 20 split-model combinations are sensitivity analyses and are not interpreted as 20 independent external replications.

4.7 Quantum-specific integrity gate

A separate simulator gate isolates the quantum-workflow boundary using CICIDS, 64/64 train/evaluation rows, five split seeds, projected dimensions 4, 6, and 8, and 11 conditions, for 165 split-dimension-condition cells. The clean negative control is compared with benign level-1 transpilation; a common post-feature-map X rewrite that must preserve fidelity; data-dependent RZ mutations at strengths 0.02 and 0.10; a feature-map repetition change from one to two; asymmetric, diagonal-eroded, and symmetric PSD-preserving kernel edits; and binomial fidelity-estimation emulators at 256 and 1,024 shots.

Each cell records canonical OpenQASM 3 circuit and parameter provenance, kernel hashes, semantic kernel differences against a trusted reference, symmetry/diagonal/eigenvalue checks, repeated-estimation discrepancy, and prediction changes. The shot conditions sample exact fidelities and therefore assess the sensor contract under controlled stochastic estimation; they are not sampler-backend or QPU executions. Acceptance checks are prespecified for the clean contract, semantic-preserving controls, circuit-mutation sensitivity, algebraic fault detection, the algebraic blind region, and finite-shot repeat disagreement.

The intended countermeasure is layered: authenticated circuit and parameter provenance, an approved-transpilation policy, semantic probe kernels, algebraic validation, repeated estimation with uncertainty bounds, PSD repair as containment rather than evidence of integrity, and output monitoring. No single sensor is treated as sufficient.

4.8 Executable service-contract demonstrator

A local HSaaS research prototype instantiates four ordered contracts: input/preprocessing, circuit/kernel, execution/result, and evaluation/report. Each contract records evidence, violations, and review conditions in an SHA-256 chain. Only four passes yield **allow**; any review yields a fail-closed **hold**, and any violation yields **block**. The chain detects later modification but is not a provider-authenticated digital signature.

The demonstrator uses one compact CICIDS cell (split 42, dimension 4, 32/32 rows) and six prespecified scenarios: clean, approved transpilation, parameterized circuit mutation, PSD-preserving kernel substitution, prior-preserving label corruption, and an approved 256-shot binomial emulator. This experiment assesses countermeasure composition and decision logic, not predictive generalization or service throughput.

5. Results

5.1 Label-path failures expose exact sensor blind regions

All random and prior-preserving evaluation-label cells have zero prediction disagreement: the model receives the same X and emits the same predictions. Feature JSD, MMD, KS rejection, and score JSD are also exactly unchanged. Balanced-accuracy conclusion impact is non-negative in the executed design, but not strictly positive in every cell. Gate 1 contains 1,276 positive-impact and 164 zero-impact label observations out of 1,440; the expansion contains 2,184 positive-impact and 1,416 zero-impact observations out of 3,600. No negative impact occurs. These are evaluation-path integrity failures when they alter the conclusion, not robustness failures of the fixed predictor.

Prior-preserving flips additionally produce zero label-prior shift and zero label JSD. They therefore occupy the exact blind region predicted by Propositions 1 and 2. In Gate 1 at flip rate 0.10, their mean conclusion impacts across evaluated model-dimension cells range from 0.0492 to 0.0633. Joint confusion evidence reacts whenever the conclusion impact is positive, showing that the observed gap closes when item-level outcomes are available and trusted.

The expansion reproduces these invariances without exception: all 3,600 evaluation-label rows are blind to feature-plus-prediction evidence and have zero prediction disagreement, and all 1,800 prior-preserving rows are also blind to label-marginal evidence. Of the 3,600 rows, 2,184 have positive conclusion impact; every one has non-zero joint-outcome evidence. Thus the coverage boundary, rather than an average detector score, replicates across all eight fixed expansion environments (Figure 1).

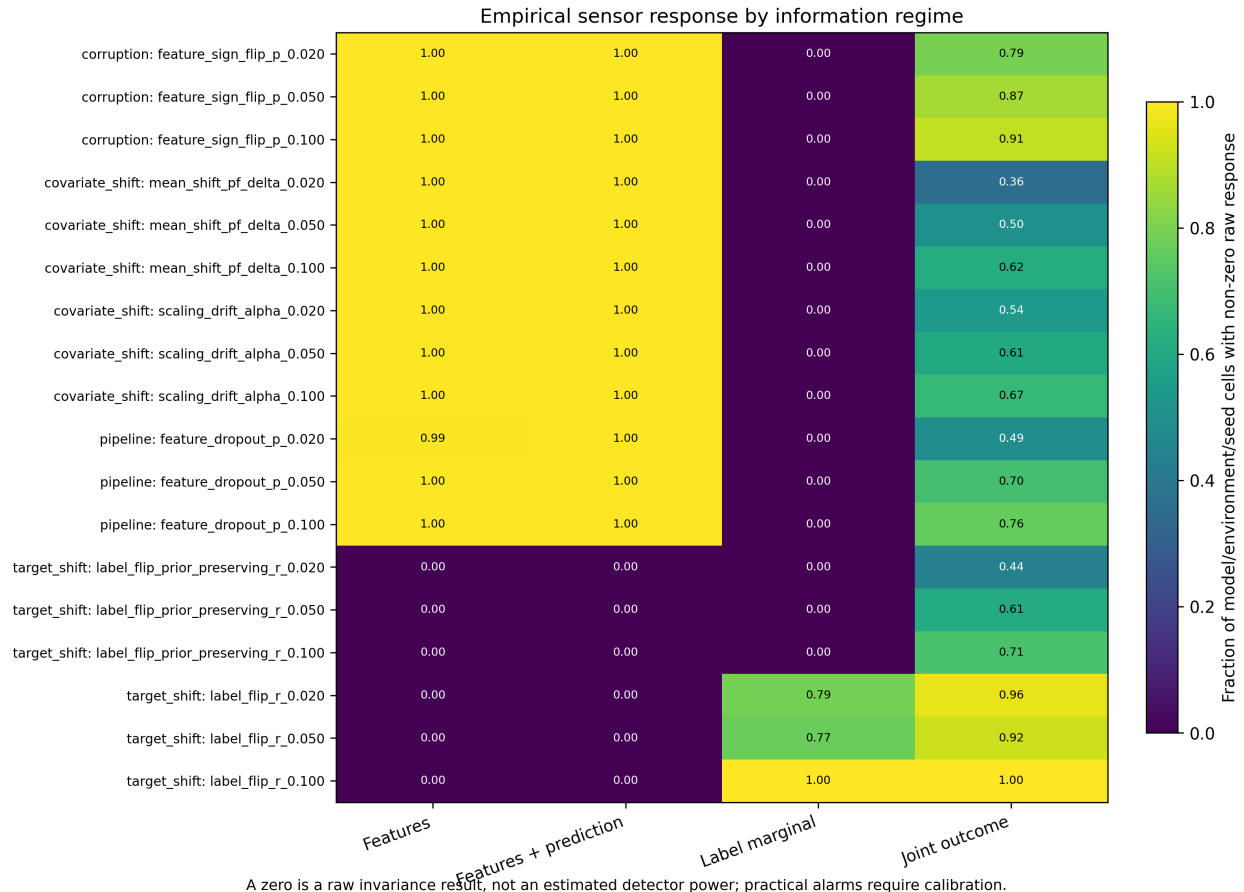


Figure 1: Empirical sensor response by information regime. Fractions are over model/environment/seed cells; zero denotes raw invariance, not estimated detector power, and practical alarms require separate calibration.

Random label flips provide a useful negative control for the information-set argument. They remain invisible to feature-plus-prediction monitoring but are strongly visible to label-marginal monitoring. The perturbation has not become intrinsically less stealthy; the auditor has acquired a more informative view.

5.2 Suite-average model impact is feature-map specific

QSVC-ZZ has the largest mean balanced-accuracy conclusion impact in the frozen ID suite. Relative to SVC-RBF, its mean ratios are 2.524, 3.094, and 3.323 at dimensions 8, 10, and 12. Ratios are descriptive; paired differences are the inferential endpoint.

The within-split paired differences are 0.02427 (95% CI 0.01200–0.03654), 0.03289 (0.01998–0.04580) and 0.03855 (0.02773–0.04937), respectively; all five split clusters are positive at each dimension. The corresponding ZZ/SVC impacts are 0.04019/0.01592, 0.04860/0.01571 and 0.05514/0.01659. These intervals summarize five split clusters in one fixed environment and are not population claims.

This is a conditional property of the frozen pipelines, dataset, suite, and dimensions. It does not establish that quantum kernels are generally more vulnerable. The other quantum profiles occupy intermediate positions

and confirm that “quantum” is not a homogeneous empirical category.

Across the eight expansion environments, the mean ZZ-minus-SVC difference is positive in 7/8 environments at dimension 8 and 8/8 at dimensions 10 and 12. Environment-specific 95% intervals exclude zero in 4/8, 7/8, and 5/8 environments, respectively. The means range from -0.00009 to 0.09382 at dimension 8, 0.00107 to 0.09508 at dimension 10, and 0.00403 to 0.11585 at dimension 12. This heterogeneity is shown in the fixed-environment forest plot in Figure 2; no pooled population effect is claimed.

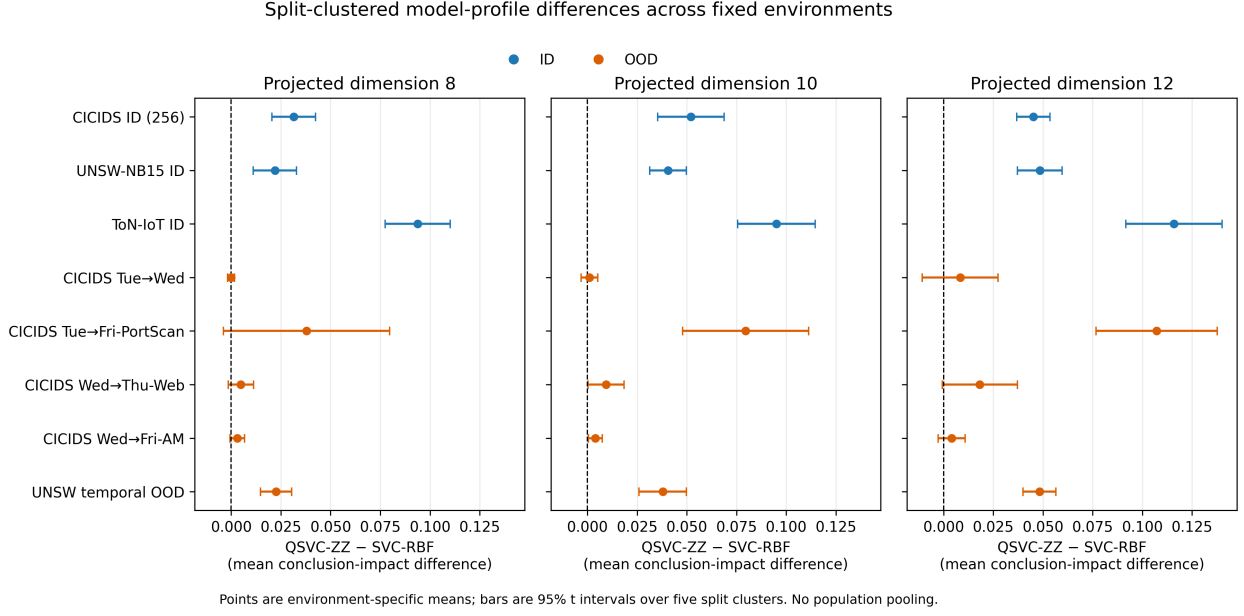


Figure 2: Environment-specific ZZ-minus-SVC mean conclusion-impact differences and 95% t intervals over five split clusters. ID and OOD settings are shown separately; no population pooling is performed.

5.3 Why a single stealth average is insufficient

An average over all normalized sensors mixes sensors that observe different information. Adding several structurally blind feature statistics can dilute one joint-outcome alarm; conversely, adding a label-aware signal assumes labels are available and trustworthy at audit time. Both change the scalar without changing the underlying evidence contract. For this reason the main result is the attack-to-sensor coverage matrix. Shared-normalized stealth remains a frozen-design sensitivity used to compare with the workshop artifact.

5.4 Cross-environment validation

All prespecified expansion gates completed with zero failed jobs. At dimension 8, seven of eight environment means for the secondary ZZ-minus-SVC profile are positive; at dimensions 10 and 12 all eight are positive. The number of environment-specific intervals excluding zero is not monotone with dimension and one OOD pair is effectively null, preventing a universal ordering claim. More importantly, the exact information-set blind regions replicate across every environment and model profile, including settings in which clean predictive performance is weak.

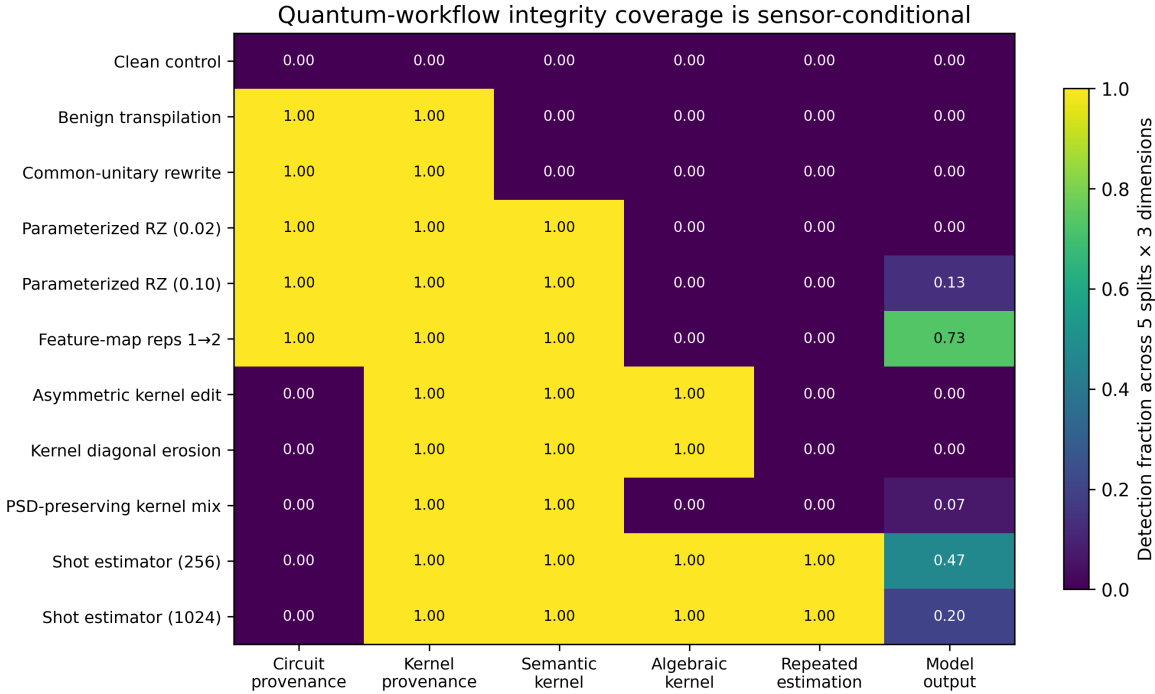
Clean balanced accuracy and suite-average conclusion impact correlate positively across expansion cells ($r=0.714$; exploratory), indicating substantial performance-headroom confounding in model-profile comparisons. The auditability result does not depend on this ordering: it is defined by invariance conditional on evidence and by whether a material conclusion change remains observable under a richer information regime.

5.5 Quantum-workflow sensor contract

All 165 prespecified cells are present and all nine acceptance checks pass under the frozen numerical tolerance ($1e-8$ for non-zero detection unless a stricter contract check is stated). Clean cells trigger no sensor. Benign transpilation and a common-unitary circuit rewrite change circuit and serialized-kernel provenance while preserving semantic kernels, algebraic properties, and outputs in every cell. They demonstrate that raw hashes require policy-aware adjudication: provenance difference alone is not evidence of harmful semantic change.

Every data-dependent circuit mutation changes the semantic kernel. The mild RZ mutation changes no evaluated prediction, the stronger mutation changes predictions in 2/15 cells, and increasing feature-map repetitions changes predictions in 11/15 cells. Conversely, every asymmetric and diagonal-eroded matrix is detected algebraically but changes no predictions at the tested severity. A PSD-preserving mixture remains symmetric, unit-diagonal, and positive semidefinite in all cells, so algebraic validation is exactly blind; trusted kernel provenance and semantic comparison detect all 15 substitutions, while output monitoring reacts in only 1/15.

Both stochastic estimators yield non-zero repeated-estimation discrepancy in all 15 cells. Output changes occur in 7/15 cells at 256 shots and 3/15 at 1,024 shots, with mean balanced-accuracy drops of 0.00521 and 0.00313. These are controlled binomial emulators, so the result supports repeated-estimation logic but not a claim about a physical device or calibrated noise model.



Provenance assumes a trusted reference; semantic and algebraic checks answer different questions. Shot rows are binomial emulation, not QPU runs.

Figure 3: Sensor-family response across clean and semantics-preserving controls, circuit mutations, kernel corruptions and finite-shot emulators. Coverage is conditional on the information exposed to each sensor; output invariance does not imply workflow integrity.

Figure 3 summarizes why these controls falsify any one-sensor security story. Circuit hashes overreact to approved semantic equivalence, algebraic checks miss valid-looking malicious substitutions, and output

monitors miss changes before they cross a decision boundary. The layered contract supplies complementary coverage and makes every residual blind region explicit.

5.6 Executable countermeasure composition

All six service-contract envelopes verify and all eight acceptance checks pass. The clean run and explicitly approved transpilation receive **allow**: despite a circuit-hash change, the transpiled semantic-kernel difference is only $7.11\text{e-}15$, below the declared tolerance. Parameterized circuit mutation and PSD-preserving kernel substitution receive **block**; the latter remains symmetric, unit-diagonal, and positive semidefinite, confirming that the block depends on trusted semantic/provenance evidence rather than algebra alone.

Prior-preserving label corruption also receives **block** at the evaluation/report contract while its class-prior delta is exactly zero. The approved 256-shot emulator receives **hold**, not **allow**: stochastic differences and repeated-estimation evidence are expected but require uncertainty adjudication. Thus the prototype evaluates an actual response policy rather than merely enumerating sensors.

The prototype is packaged with a console entry point, tests, CI configuration, a strict output manifest, and reviewer-facing summary table. It remains a local process and makes no claim about network authentication, signed provider records, persistence, incident handling, scheduling or QPU binding.

6. Integrity-audit contract for hybrid workflows

Every integrity claim should report seven fields:

1. **Boundary:** acquisition, preprocessing, circuit construction, transpilation, execution, kernel estimation, prediction, label, or reporting.
2. **Protected asset:** data, circuit IP, execution fidelity, kernel matrix, prediction, ground truth, or reported conclusion.
3. **Adversary or failure capability:** exactly which elements may change and which remain fixed.
4. **Information regime:** which of I_X , I_{XF} , I_{Ym} , I_{XFY} , and I_Q is available.
5. **Sensor and calibration:** raw statistic, null reference, threshold, and false-alarm budget.
6. **Coverage:** perturbations for which the sensor can react, exact blind regions, and empirical power where estimable.
7. **Response:** provenance failure, abstention, rerun, PSD repair, rollback, human adjudication, or another countermeasure.

For label-path integrity, the minimum defensive bundle is item-level label provenance plus a joint label-prediction audit. The simulator gate evaluates the corresponding bounded quantum-kernel bundle: circuit and parameter provenance, transpilation-aware adjudication, semantic probes, kernel symmetry/diagonal/PSD checks, repeated-estimation discrepancies, and a fail-closed policy when evidence violates the contract. These controls specify the evidence required inside the studied boundary; they do not certify any provider or hardware boundary.

7. Discussion

7.1 Robustness and integrity are causally different

The label-path result is intentionally simple because it exposes a common category error. A performance metric can fall even though the model output is bit-for-bit unchanged. Calling that event model degradation assigns the failure to the wrong component and suggests the wrong defense. Input robustness cannot repair corrupted ground truth; provenance and joint validation can.

7.2 Auditability claims must name their information set

Zero response is evidence only when a sensor had access to information capable of distinguishing the states. This principle applies beyond label corruption. A circuit hash cannot detect a calibration change, a kernel

PSD check cannot establish label provenance, and an output monitor cannot protect circuit confidentiality. Hybrid security therefore needs a coverage argument across boundaries rather than one global anomaly score.

7.3 The quantum result is a profile, not a universal ordering

QSVC-ZZ’s larger suite-average impact is directionally consistent in most fixed environments, but its magnitude and interval support vary sharply across ID/OOD pairs. Clean-performance headroom, pipeline-specific scaling, and kernel geometry are entangled. A causal attribution to quantum geometry would require symmetric preprocessing controls, headroom-aware endpoints, kernel-level interventions, or a decomposition experiment. Until then the paper reports feature-map-specific profiles and avoids universal classical-versus-quantum vulnerability language.

7.4 Relationship to ATHENA-AEGIS

The framework provides direct evidence for ATHENA-AEGIS G3.2 (validation and verification across the studied hybrid steps) and G3.3 (metrics, threat identification and enforcement), and selected simulator evidence for G3.1. It covers classical preprocessing, prediction, outcome and result interpretation; adds bounded evidence for circuit construction, benign transpilation equivalence, controlled kernel corruption and stochastic fidelity estimation; and executes the resulting policy through a four-contract local prototype. The software is therefore a research contribution towards Result 3.1 and the manuscript supports Result 3.2; neither result is claimed complete for the whole project.

Within WP5, a submitted or published article contributes to Task 5.2 and its scientific-dissemination report. This paper does not implement Task 5.1 or Result 5.1, the operational Fleet Management case study. Dataset choice and the local HSaaS demonstrator must not be used to imply that operational use-case coverage.

7.5 What the integrated contribution adds

The closest precedents each cover an important portion of the design. QCIVET already supplies stage contracts, hash-linked traces and stronger backend/QPU evidence [YeniarasKarimov2026]; evidence-sufficiency monitoring formalizes limitations of proxies under delayed outcomes [SoloZobov2026]; circuit-integrity metrics and quantum software testing establish complementary quantum-side checks [Ahmed2026, Wang2021, Pal-tenghi2023, Luo2026]. This paper’s additional claim is narrower but joint: the audit conclusion is indexed by the exact information exposed, the evaluation-label path is a first-class protected boundary, structural blind regions are separated from empirical response, the coverage boundary is replayed across fixed dataset/OOD environments, and the same coverage map drives a fail-closed local decision. None of the propositions, contracts, hashes, sensors or model comparisons is presented as sufficient novelty alone.

8. Limitations

First, the expansion includes three datasets and multiple temporal pairs, but these are eight fixed cybersecurity environments assembled from repository subsets. They improve transport evidence but are not a random sample of deployments, and five split clusters quantify within-environment sampling rather than population-level generalization.

Second, most gates cap training and evaluation at 128; the completed 256/256 CICIDS gate is a scale sensitivity, not evidence of large-scale quantum-kernel deployment.

Third, both the archived Qiskit reference evaluator and the accelerated exact-statevector evaluator operate in ideal simulation. The quantum-specific gate adds controlled binomial shot estimation, but not a sampler backend, device-calibrated noise, or hardware experiment.

Fourth, SVC and QSVC use their frozen pipeline-specific scalars. The study compares complete pipelines; it does not causally isolate preprocessing from kernel geometry.

Fifth, the quantum-specific gate covers controlled circuit/parameter mutation, benign transpilation equivalence, post-estimation matrix corruption, and finite-shot emulation. It does not cover malicious compiler

or scheduler attacks, calibrated hardware noise, provider manipulation, multi-tenant interference, or side channels.

Sixth, exact blindness is a capability statement under the specified perturbation. It does not estimate how often such failures occur in deployment or imply malicious intent.

Seventh, min-max stealth is reference-dependent and retained only as a secondary sensitivity. A practical alarm requires null calibration, false-alarm control, and detection-power estimates.

Eighth, the HSaaS demonstrator is an executable local contract, not a deployed network service or an authenticated provider ledger. Its hash chain is tamper-evident but supplies neither identity nor non-repudiation. The article does not evaluate Fleet Management, scheduling, provider security, multi-tenancy or physical quantum execution; those topics are excluded rather than deferred experimental claims of this study.

9. Conclusion

Hybrid workflow integrity cannot be reduced to a robustness number. A perturbation can change a reported conclusion without changing the model, and a detector can remain silent because the evidence it sees is mathematically invariant. We formalized this dependency through information-set conditional auditability, proved exact blind regions for label-only and prior-preserving perturbations, and reproduced those boundaries across CICIDS, UNSW-NB15, ToN-IoT, and eight fixed expansion environments. The feature-map-specific QSVC-ZZ impact profile is directionally consistent in most environments but heterogeneous and headroom-confounded, so it remains secondary and conditional.

The methodological requirement is straightforward: every integrity claim must state the workflow boundary, information available, sensor coverage, residual blind region and response. This turns auditability from an unqualified scalar into a bounded evidence contract. The data-to-evaluation layer has multi-dataset, temporal, scale and clustered evidence; the quantum-specific simulator layer validates complementary controls; and the packaged HSaaS prototype turns those controls into fail-closed decisions. The resulting claim is complete for the declared research boundary and intentionally makes no full-lifecycle, provider, Fleet Management or hardware-security claim.

Declarations

Funding. Funding details are supplied on the separate title page and are withheld from this file for double-anonymized review.

Data and code availability. The review artifact contains the source code, locked environment specification, tests, compact derived tables, figures, SHA-256 manifests and exact reproduction commands. Raw CICIDS2017, UNSW-NB15 and ToN-IoT data are not redistributed; the documentation records their public sources and deterministic staging steps. During double-anonymized review the archive is supplied without author-identifying metadata. The version DOI is to be inserted in the unblinded record after archival.

Ethics. The study uses previously released network-security benchmark data and performs no research involving human participants, animals or newly collected personal data.

Competing interests and author contributions. These declarations are kept on the separate title page for double-anonymized review and require confirmation by all authors before upload.

Declaration of generative AI and AI-assisted technologies. During manuscript and artifact preparation, the authors used OpenAI Codex for literature-discovery assistance, language editing, code inspection and test automation. The authors verified the cited primary sources, numerical claims and generated edits and retain full responsibility for the article.

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